LLM-assisted voxel synthesis

From reusable programs to diverse source shapes

1 Specification

SPEC

21 classes · 40 families each

48^3 grid · 750–1,250 voxels

2 LLM programs

```
AI_A + AI_B
shape(params)
emit(x, y, z)
```

Write + run generation code

Deterministic, parameterized shapes

3 Voxels + captions

L1–L5

Voxel models + descriptions

Connectivity + semantic checks

4 Deduplication

Exact + canonical hashing

48 axis transforms · 80k unique

If needed: repair / retile

5 Tiling + repair

Layer-wise exact cover

MRV / MILP / beam · ≤2% edits

6 Contact screening

mg

Self-weight LP + layer prefixes

Tipping margin + clutch force

7 Training records

LDR TXT JSON

Brick layouts + 5 captions

Round-trip audit + screening status

8 BrickMore model

LoRA → Qwen3-1.7B

Text → brick sequence + repair

Voxel-to-brick conversion, screening and downstream training